

SIMATS ENGINEERING

ASSESSMENT TOOL 2: Concept Mapping

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1519

COURSE OUTCOME COVERED

CO1: *Apply cloud computing technologies including hardware-software evolution, virtualization, web services, and service models to develop cloud-based solutions. (BL2)*

Assessment Tool Weightage: 20%

Dave's Taxonomy: Imitation (Level 1)
Question Count: 4

Total Marks: 20

Code of Conduct

I **Jannathul Firdaus. N (192573025)** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.



Signature of the student:

Assessment Tasks

Concept Mapping Tasks

Create concept maps for the following tasks. Each map must show key nodes, link labels, and a logical hierarchy.

1. Concept Map 1: Cloud Computing Fundamentals. Include definition, characteristics, benefits, and limitations.
2. Concept Map 2: Evolution of Cloud Computing. Connect hardware evolution, internet software evolution, distributed systems, and service delivery.
3. Concept Map 3: Service Models. Map IaaS, PaaS, SaaS, and XaaS with provider responsibilities and user responsibilities.
4. Concept Map 4: Virtualization and Web Services as Enablers of Cloud. Show how these two concepts support cloud deployment and scalability.

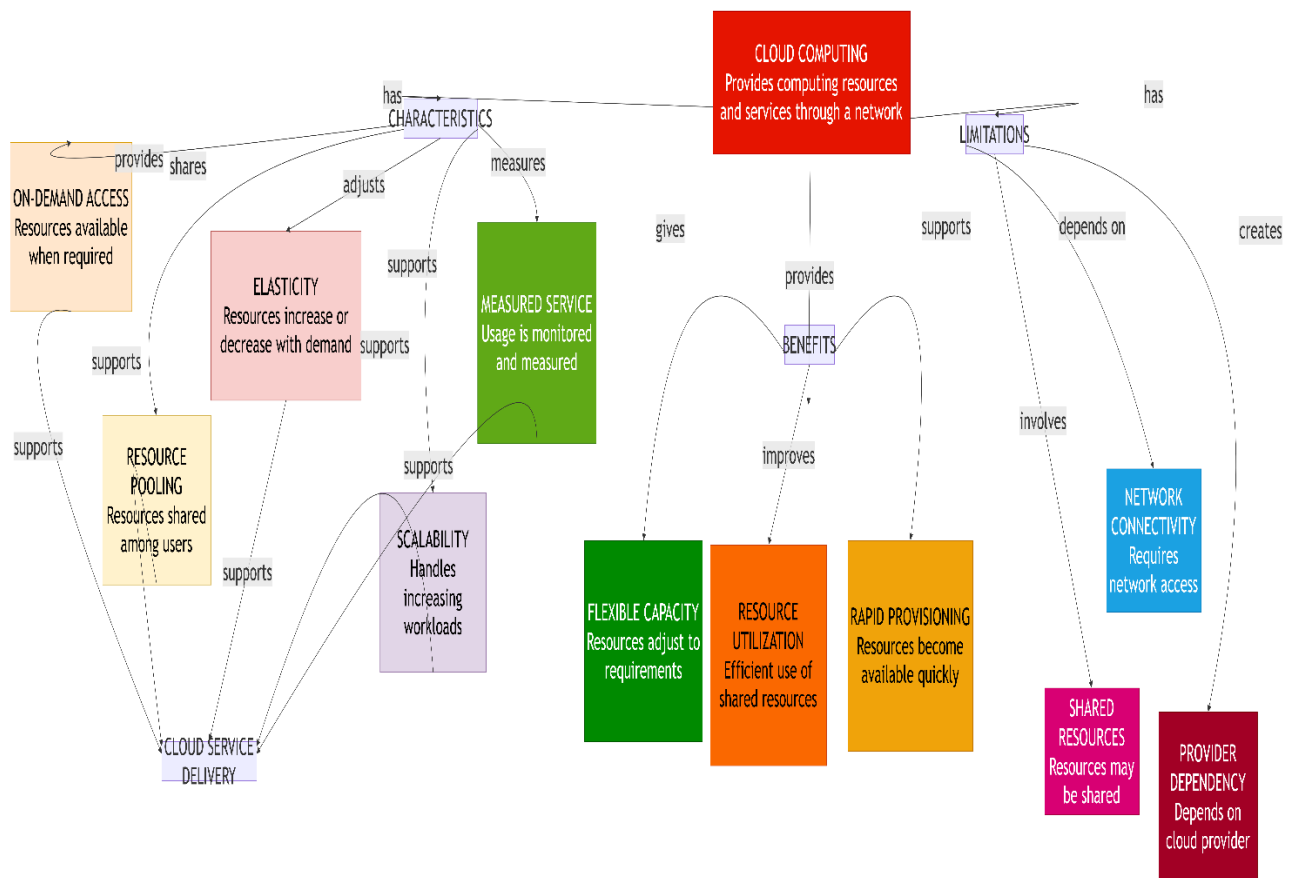
Expected concept nodes:

- Elasticity, scalability, resource pooling, on-demand access, measured service
- Virtual machine, hypervisor, abstraction, API, SOAP, REST, provider, consumer
- Infrastructure layer, platform layer, application layer, shared responsibility

Concept Mapping Rubric

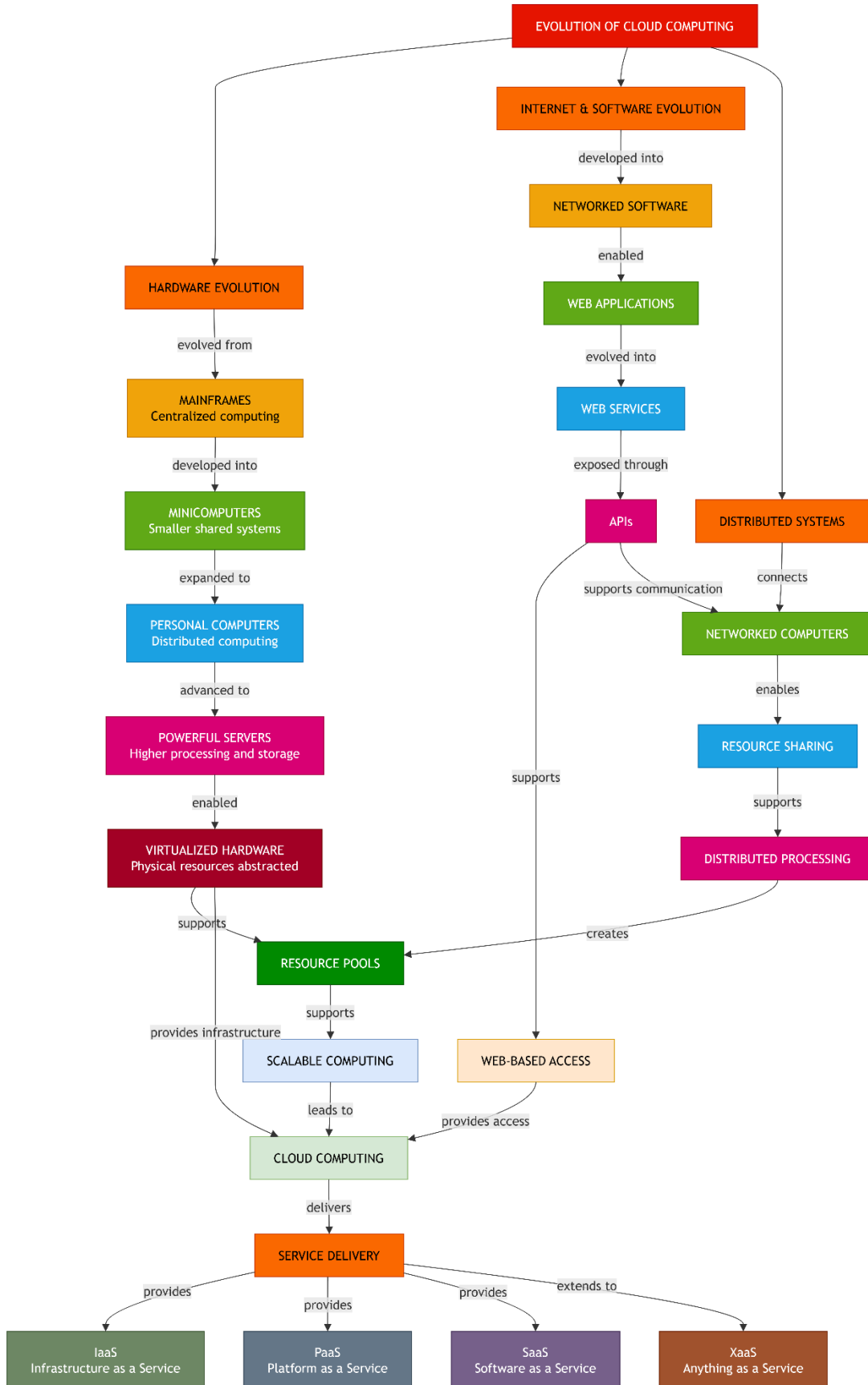
Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Concept Identification	25%	All major concepts are accurately identified	Most concepts are identified correctly	Limited concepts are identified	Essential concepts are missing
Relationship Mapping	25%	Links between concepts are accurate and meaningful	Most links are correct	Some links are weak or unclear	Relationships are mostly incorrect
Hierarchy and Organization	20%	Excellent structure with clear hierarchy	Mostly organized with minor issues	Basic structure with weak hierarchy	No logical organization
Technical Relevance	20%	Map strongly reflects cloud course content	Mostly aligned to topic	Partially aligned to topic	Weak alignment to topic
Visual Clarity	10%	Neat, readable, and easy to interpret	Generally readable	Somewhat cluttered	Poorly presented

1. Cloud Computing Fundamentals



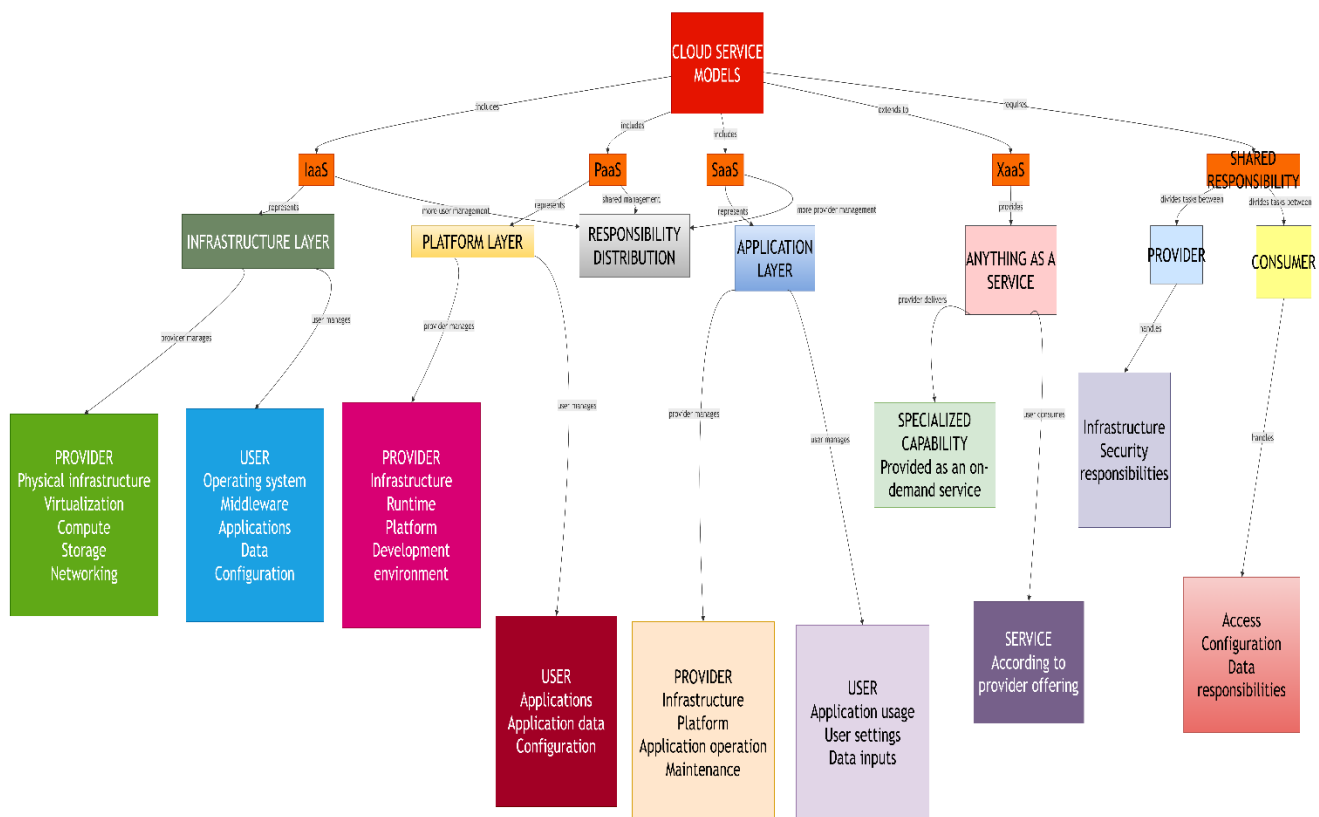
- **Cloud computing** provides computing resources and services through network-based access.
- **On-demand access** allows users to obtain resources whenever required.
- **Resource pooling** enables shared use of provider resources.
- **Elasticity and scalability** allow cloud resources to respond to changing workloads.
- **Measured service** enables monitoring and measurement of resource usage.
- Benefits include **flexibility, efficient utilization, and rapid provisioning**.
- Limitations include **network dependence, shared-resource concerns, and provider dependency**.

2. Evolution of Cloud Computing



- **Hardware evolution** moved from centralized mainframes toward powerful servers and virtualized hardware.
- **Internet and software evolution** introduced web applications, web services, APIs, and browser-based access.
- **Distributed systems** connected computers and enabled resource sharing and distributed processing.
- These developments contributed to **resource pooling and scalable computing**.
- Cloud computing combines these developments to provide **service delivery**.
- Service delivery includes **IaaS, PaaS, SaaS, and XaaS**

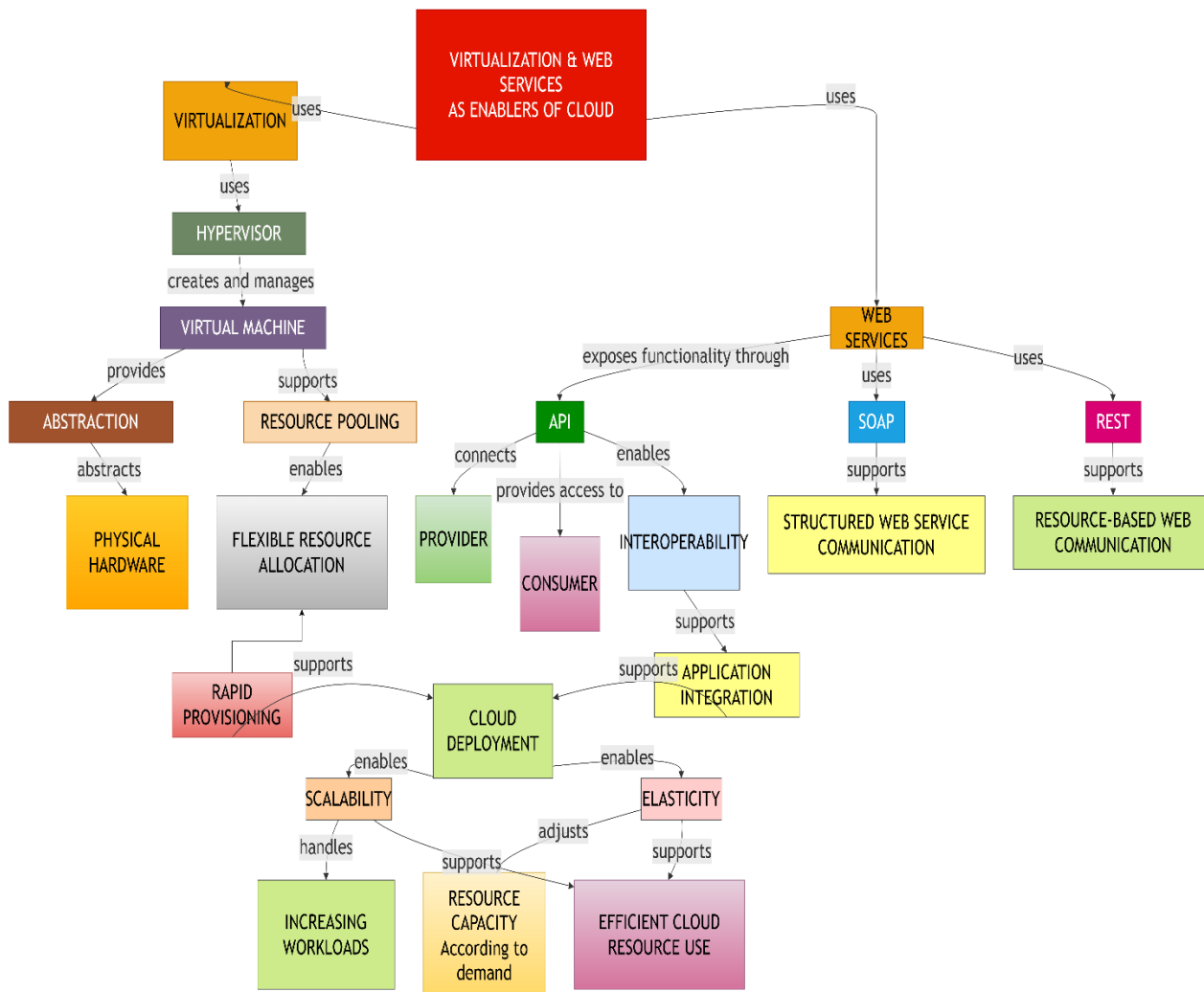
3. Cloud Service Models



- **IaaS** provides infrastructure resources. The user manages the operating system, applications, and data.
- **PaaS** provides a platform and development environment. The provider manages more of the underlying infrastructure.
- **SaaS** provides complete applications. The provider manages most of the application stack.
- **XaaS** represents specialized capabilities delivered as services.
- Across all models, there is a **shared responsibility** between provider and consumer.

- Moving from **IaaS** → **PaaS** → **SaaS**, more infrastructure and platform responsibilities are handled by the provider.

4. Virtualization and Web Services as Enablers of Cloud



- **Virtualization** abstracts physical hardware and allows resources to be divided into virtual environments.
- A **hypervisor** creates and manages virtual machines.
- **Virtual machines** support resource pooling and flexible allocation.
- Flexible allocation enables **rapid provisioning** of cloud resources.
- **Web services** expose functionality through standardized interfaces.
- An **API** provides communication between the cloud provider and consumer.
- **SOAP** supports structured web-service communication.

- **REST** provides resource-based web communication.
- APIs improve **interoperability and application integration**.
- Virtualization and web services together support **cloud deployment, scalability, and elasticity**.